

The Effect of pH on the Allosteric Behaviour of Ox-Brain NAD^+ -Dependent Isocitrate Dehydrogenase

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The sigmoid dependence of the initial rate of the reaction catalysed by NAD^+ -dependent isocitrate dehydrogenase on the concentration of isocitrate is greatly reduced as the pH is decreased. The apparent pK for this process is 7.3. At pH values of 6.0 and 6.5 the enzyme exhibits hyperbolic kinetics with respect to isocitrate at relatively high concentrations of NAD^+ , but the dependence becomes sigmoid at lower NAD^+ concentrations. The allosteric activators ADP and citrate have only small effects on the activity of the enzyme at pH 6.5 but in the latter case the effects are increased by decreasing the concentration of NAD^+ . The dependence of initial velocity on the NAD^+ concentration is hyperbolic at all pH values in the pH range 6–8.5. NADH is a competitive inhibitor of enzyme activity with respect to NAD^+ , and its presence induces a sigmoid dependence of initial velocity on isocitrate concentration at pH 6.5 and in the presence of high concentrations of NAD^+ . Kinetic studies at pH 6.5 indicate that the apparent maximum velocity of the reaction with respect to NAD^+ is insensitive to changes in the isocitrate concentration and that with isocitrate as the substrate is relatively insensitive to changes in the NAD^+ concentration under conditions where the behaviour appears to be hyperbolic. *threo*- D_5 -Isocitrate is the true substrate for the enzyme and the corresponding L_5 isomer is neither a substrate nor an inhibitor.

Studies on the allosteric properties of ox brain NAD^+ -dependent isocitrate dehydrogenase at pH 7.5 [1] have shown that, in common with the enzyme from a number of other mammalian sources (see [2] for example), the initial velocity of the reaction catalysed depends hyperbolically on the concentration of NAD^+ but sigmoidally on the concentration of isocitrate. The enzyme from pig liver has, however, been reported to respond hyperbolically to changes in the isocitrate concentration [3]. This difference could have resulted from the lower pH of 7.2 used in the latter studies since Garland and Nichols [4] have reported that the enzyme from rat liver shows allosteric behaviour at pH 7.5 but not at pH 7.0.

Changes in the assay pH have been shown to cause a number of enzymes to lose their allosteric properties (see [5] for example) and studies under such conditions can be used to obtain information on the kinetic mechanism followed. Although the possibility that alterations in the pH may affect the kinetic mechanism

cannot be excluded, such investigations can yield information of potential value to interpretation of the function of the enzyme that could not be obtained so readily under conditions where apparent homotropic cooperativity existed.

A study of the effects of pH on the allosteric properties NAD^+ -dependent isocitrate dehydrogenase could be useful in determining whether the reported difference between the behaviours of pig liver enzyme and those from other mammalian sources simply arose because of differences in the assay conditions and could also offer the possibility of studying the kinetics of the enzyme under conditions where its allosteric properties had been lost. This paper reports the results of such a study.

MATERIALS AND METHODS

The NAD^+ -dependent isocitrate dehydrogenase was purified from ox brain by the method previously described [6]. Coenzymes, ADP and triethanolamine chloride were obtained from C. F. Boehringer GmbH (Mannheim, FRG). Tris base and *threo*- D_5 -isocitrate

Abbreviation. $s_{0.5}$, substrate concentration necessary to give half maximum velocity.

Enzyme. NAD^+ -dependent isocitrate dehydrogenase or *threo*- D_5 -isocitrate: NAD^+ oxidoreductase (decarboxylating) (EC 1.1.1.41).

(monopotassium salt) were from Sigma Chemical Company (London, England). Sephadex G-25 was obtained from Pharmacia (Uppsala, Sweden). All other chemicals were obtained from B.D.H. Ltd (Poole, Dorset, England).

The concentrations of solutions of ADP and NADH were determined spectrophotometrically taking the absorption coefficient of ATP at 260 nm to be $13400 \text{ M}^{-1} \text{ cm}^{-1}$ and that for NADH at 340 nm to be $6330 \text{ M}^{-1} \text{ cm}^{-1}$. Solutions were stored at 4°C and were used within three days of preparation.

The activity of NAD^+ -dependent isocitrate dehydrogenase was determined spectrophotometrically at 25°C with a Unicam SP 1800 spectrophotometer fitted with a Unicam AR 25 linear recorder. The compositions of the assay mixtures used, which each had a volume of 1.0 ml, are given in the appropriate figures. A racemic *threo*- D_sL_s -isocitrate preparation was used in the majority of these studies, but since the L_s isomer is neither a substrate nor an inhibitor of the enzyme used (see later), the isocitrate concentrations are expressed in terms of the D_s component. One unit (U) of enzyme activity is defined as the amount catalysing the formation of one micromole of NADH in one minute.

Kinetic data that obeyed the Michaelis-Menten equation were fitted by weighted regression analysis using a computer program similar to that described by Cleland [7]. A similar program was used to fit data for reversible inhibition. In the case of results that gave rise to a sigmoid dependence of initial velocity on substrate concentration the data were fitted by eye to the linear transformation of the Hill equation [8]:

$$\log \frac{v}{V-v} = h \log s - \log K$$

where v is the initial velocity at substrate concentration s , V is the maximum velocity, h is the Hill coefficient and K is related to the substrate concentration necessary to give half-maximum velocity ($s_{0.5}$) by the function:

$$s_{0.5} = h \sqrt{K}.$$

RESULTS

The kinetics of the reaction catalysed were studied over a range of pH values. The stability of the enzyme at these pH values was determined by adjusting samples to various values and incubating at 25°C for 30 min before they were transferred to 100 mM triethanolamine chloride buffer, pH 7.5 for assay. Full activity was regained after such treatment over the pH range 5.5–8.5 indicating the effects on enzyme activity to be fully reversible. Variation of the ionic strength over the range that would occur

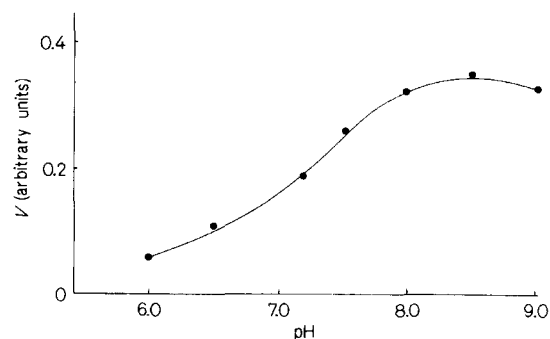


Fig. 1. The effect of pH on the maximum velocity of the reaction catalysed by NAD^+ -dependent isocitrate dehydrogenase. The assay mixtures contained 1 mM NAD^+ , 10 mM MgSO_4 enzyme and buffer. The buffers used were: 100 mM imidazole-HCl, pH 6.0 and 6.5; 100 mM triethanolamine chloride, pH 7.2–9.0. The maximum velocities with respect to isocitrate were obtained by extrapolation of double-reciprocal plots

with the different buffer systems used, by the addition of NaCl, had no effect on the activity of the enzyme and therefore no attempt was made to maintain this constant. The initial velocity was also found to be proportional to enzyme concentration over this range.

The effects of pH on the activity of the enzyme determined at high concentrations of its substrates and magnesium ions is shown in Fig. 1. The optimum pH under these conditions is at about 8.4 and there appears to be no significant activatory or inhibitory effect of any of the buffers used. Because of low activities it was not found possible to carry out accurate kinetic studies outside the pH range 6–8.5. The results of initial rate studies at the extrema of this range in the presence of high concentrations of NAD^+ and MgSO_4 and in the presence and absence of the allosteric activator ADP are shown in Fig. 2. These results indicate that the markedly sigmoid response to isocitrate concentration is lost at lower pH values.

Analysis of the dependence of initial velocity on the concentration of isocitrate according to the Hill equation allowed the Hill coefficient to be determined at several different pH values. Fig. 3 shows the results of these studies superimposed on a theoretical titration curve for a group with a pK of 7.3. From these results it can be seen that the enzyme did not show apparent homotropic cooperativity below a pH of about 7.0, the Hill coefficient approximating to unity. The agreement between the experimental points and the theoretical curve suggests that a group with an apparent pK of 7.3 may be involved in the expression of homotropic cooperativity under these conditions. The effect of pH on the substrate concentration necessary to give half-maximum velocity ($s_{0.5}$) is shown in Fig. 4.

Since the enzyme appeared to exhibit hyperbolic kinetics below pH 7, studies were carried out at pH 6.5 to determine the stereochemistry of the substrate

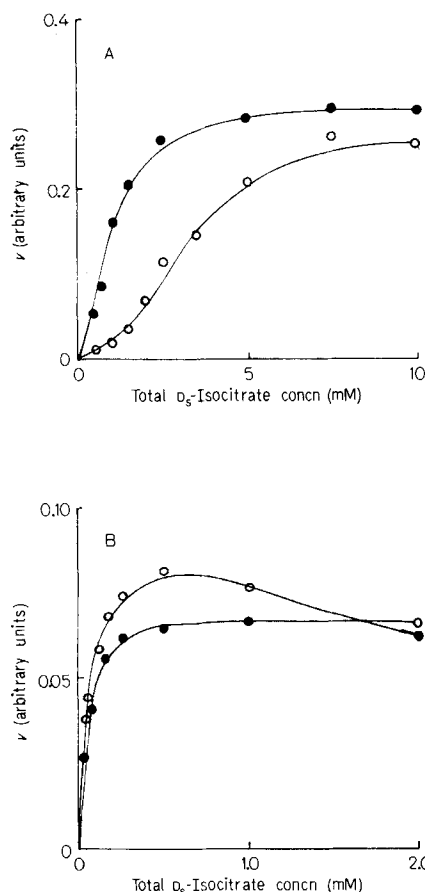


Fig. 2. Dependence of initial velocity on the total D_s -isocitrate concentration. The assay mixtures contained 1 mM NAD^+ , 10 mM $MgSO_4$, enzyme, isocitrate and buffer. 2 mM ADP was either absent (○) or present (●). (A) Assayed in 100 mM triethanolamine chloride buffer, pH 8.5. (B) Assayed in 100 mM imidazole-HCl buffer, pH 6.0

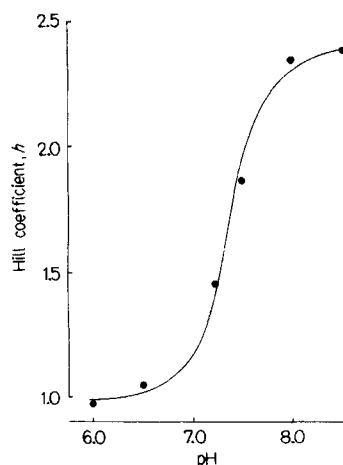


Fig. 3. The effect of pH on the Hill coefficient for isocitrate. The experimental points were determined from Hill plots of data obtained under the experimental conditions described in Fig. 2 using 100 mM imidazole-HCl buffer for pH of 6.0 and 6.5 and 100 mM triethanolamine chloride buffer for the other values. The solid line is a theoretical titration curve for a group with a pK of 7.3

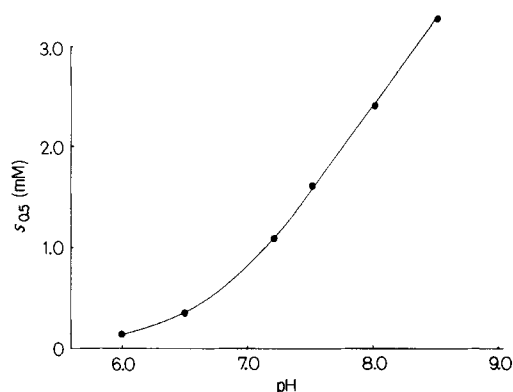


Fig. 4. The effect of pH on the D_s -isocitrate concentration necessary to give half-maximal velocity ($s_{0.5}$). Values were determined as described in Fig. 3

that is acted upon by the ox brain enzyme (see [9] for nomenclature). A comparison of the kinetics of isocitrate oxidation in the presence of 1 mM NAD^+ and 10 mM $MgSO_4$ revealed no significant difference between the K_m and V values given when the pure *threo*- D_s -isomer and the racemic *threo*- D_s - L_s mixture were used when the concentration of the latter was expressed in terms of its *threo*- D_s isomer content, indicating that the L_s isomer is neither a substrate nor a significant inhibitor of the enzyme.

The response of initial velocity to NAD^+ concentration, determined in the presence of high concentrations of isocitrate and $MgSO_4$, was hyperbolic at pH values of 6.0, 8.0 and 8.5 with K_m values of 0.061 mM (pH 6), 0.36 mM (pH 8) and 4.34 mM (pH 8.5). The effects of different concentrations of isocitrate on the initial rate of NAD^+ reduction at pH 6.5 are shown in double-reciprocal form in Fig. 5A. The lines obtained appeared to intersect on the reciprocal-velocity axis, and a replot of their slopes against the reciprocal of the isocitrate concentration (Fig. 5B) was linear.

Double-reciprocal plots of the variation of initial velocity with isocitrate concentration at a number of fixed concentrations of NAD^+ are shown in Fig. 6. At higher NAD^+ concentrations the plots appeared linear and to intersect on the reciprocal-velocity axis. However the lines obtained at lower concentrations of NAD^+ became curved, indicating that the apparent cooperativity returns at lower NAD^+ concentrations; similar results were obtained at pH 6.0. Hill plots of the data in Fig. 6 showed that the Hill coefficient increased from 1.0 at 1 mM NAD^+ to 1.5 at 0.1 mM NAD^+ .

At pH 7.5 $NADH$ was shown to be a competitive inhibitor with respect to NAD^+ but the sigmoid response to isocitrate prevented a study of the type of inhibition towards this substrate [1]. An attempt to study the inhibition at pH 6.5 revealed that the presence of $NADH$ resulted in the response to isocitrate

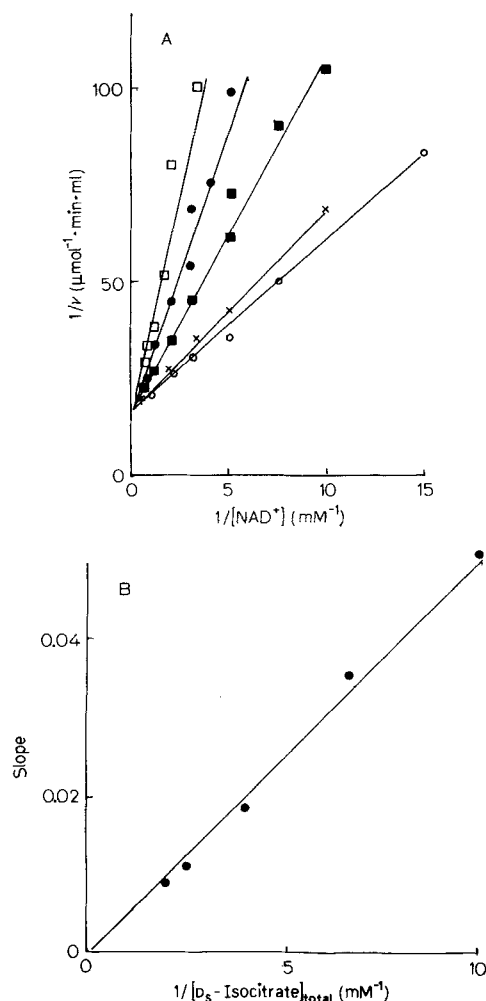


Fig. 5. Variation of the initial velocity of the reaction catalysed by NAD^+ -dependent isocitrate dehydrogenase with the concentration of NAD^+ . The assay mixtures contained 10 mM MgSO_4 , 100 mM imidazole-HCl buffer, pH 6.5, enzyme, isocitrate and NAD^+ . (A) Double-reciprocal plots at the following total concentrations of D_s -isocitrate: ($\square$) 0.1 mM; ($\bullet$) 0.15 mM; ($\blacksquare$) 0.25 mM; ($\times$) 0.4 mM; ($\circ$) 0.5 mM. (B) Replots of the slopes of the lines in (A) against the reciprocal of the D_s -isocitrate concentration

concentration changing from apparently hyperbolic to sigmoid kinetics (Fig. 7).

No inhibition by the product 2-oxoglutarate, at concentrations of up to 8 mM, could be detected at either pH 6.5 or pH 7.5 with concentrations of NAD^+ and isocitrate in the ranges 0.1–1 mM and 0.2–4 mM respectively. No oxidation of 0.1 mM NADH could be detected at either pH 6.0 or pH 7.5 in reaction mixtures containing 10 mM MgSO_4 , 5 mM 2-oxoglutarate and enzyme that were saturated with CO_2 or were mixed with 50 mM NaHCO_3 .

The effects of the allosteric activator ADP on the activity of the enzyme at pH 6.5 and 7.5 are compared in Fig. 8. There was still a weak activation at pH 6.5, confirming that the allosteric effects had not been completely abolished by lowering the pH. At pH 6.0,

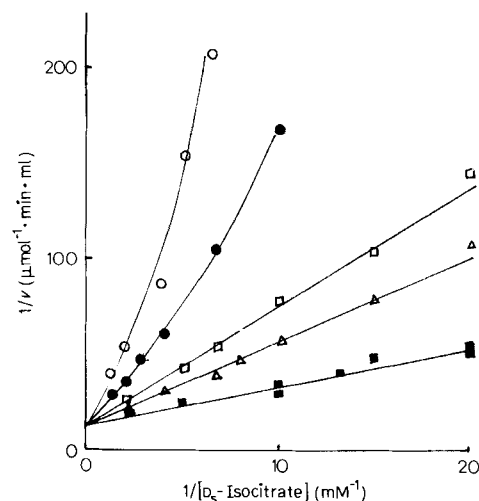


Fig. 6. Variation of the initial velocity of the reaction catalysed by NAD^+ -dependent isocitrate dehydrogenase with the concentration of D_s -isocitrate. Assay mixtures were as described in Fig. 5. The NAD^+ concentrations used were: ($\blacksquare$) 1.0 mM; ($\blacktriangle$) 0.5 mM; ($\square$) 0.3 mM; ($\bullet$) 0.2 mM; ($\circ$) 0.1 mM

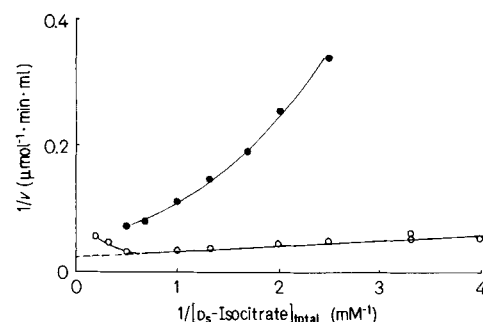


Fig. 7. The effects of NADH on the oxidation of isocitrate by NAD^+ -dependent isocitrate dehydrogenase at pH 6.5. The reaction mixture contained 1 mM NAD^+ , 10 mM MgSO_4 , 100 mM imidazole-HCl buffer, pH 6.5, enzyme and isocitrate. 50 μM NADH was present in the assays designated by ($\bullet$) and absent from those designated ($\circ$)

however, the allosteric effects were not observed and, in fact, a slight inhibitory effect was observed (Fig. 2A). This effect could perhaps have been due to catalytically unproductive binding of ADP to the NAD^+ sites on the enzyme, particularly since ADP was present in a twofold excess over NAD^+ . At all pH values studied, ADP had little effect on the maximum velocity and the Hill coefficient but did affect the value of $s_{0.5}$. Citrate is also an activator of the enzyme from this source [1] as well as those from yeast [10] and plants [11], but it has been reported not to affect the activity of the enzyme from pig liver [3]. Fig. 9 shows that the activatory effect at pH 6.5 is dependent on the concentration of NAD^+ , being much more pronounced at concentrations close to the K_m value.

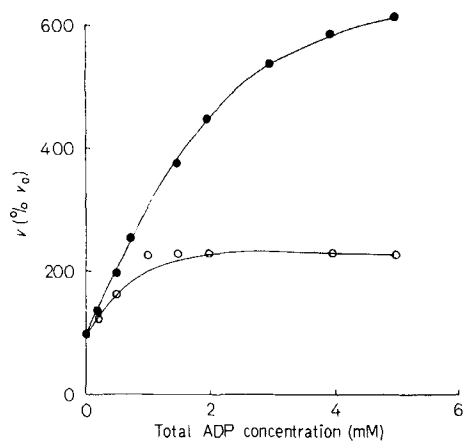


Fig. 8. The effects of ADP on the initial rate of the reaction catalysed by NAD^+ -dependent isocitrate dehydrogenase. The reaction mixture contained 1 mM NAD^+ , 10 mM MgSO_4 , 1.5 mM D_3 -isocitrate, enzyme and buffer. The buffers used were: (●) 100 mM triethanolamine chloride, pH 7.5, and (○) 100 mM imidazole-HCl buffer, pH 6.5. The initial velocity is expressed as a percentage of that in the absence of ADP at that pH value (v_0)

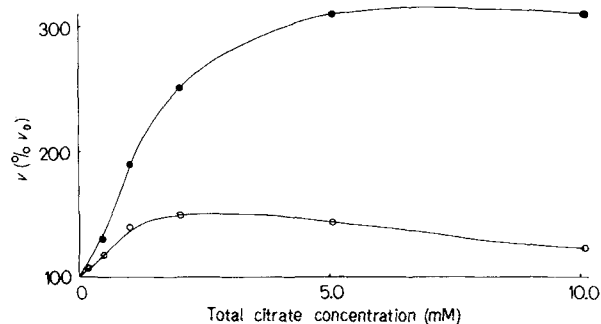


Fig. 9. The effect of citrate on the initial rate of the reaction catalysed by NAD^+ -dependent isocitrate dehydrogenase at pH 6.5. The assay mixtures contained 10 mM MgSO_4 , 100 mM imidazole-HCl buffer, pH 6.5, 0.15 mM D_3 -isocitrate, enzyme and citrate. Assays designated (○) contained 1.0 mM NAD^+ and those designated (●) contained 0.2 mM NAD^+ . The initial velocity is expressed as a percentage of that in the absence of citrate (v_0)

DISCUSSION

The effects of pH upon the allosteric properties of NAD^+ -dependent isocitrate dehydrogenase may explain some of the reported differences between the behaviours of the enzyme from different sources. The loss of the sigmoid dependence of initial velocity upon isocitrate concentration that occurs as the pH is lowered, and when a high concentration of NAD^+ is present in the assay mixture, could account for the failure of Plaut et al. [3] to observe any response with the enzyme from pig liver since the effect would be quite small at the assay pH of 7.2 used in that work (see Fig. 3). The reported insensitivity of the pig liver enzyme to activation by citrate [3] may also result from the lower assay pH used in those studies since

Fig. 9 shows that the activation is small at lower pH values if the concentration of NAD^+ is high. A difference between the behaviour of the ox brain enzyme and that from pig heart which is less easy to understand concerns the effect of pH on the K_m for NAD^+ since in the latter case this has been reported not to vary with pH [13].

The dependence of the Hill coefficient for isocitrate on pH suggests that a group with an apparent pK of 7.3 may be involved in the expression of cooperativity. Because lowering the pH value to 6.0 does not result in complete loss of allosteric properties, analysis of the effects of pH on the Michaelis parameters and an $s_{0.5}$ in terms of the theories that have been developed for simpler systems (see [12] for example) may have little validity. Colman [2] has identified a group with a pK of 6.6, which was subsequently identified as a carboxyl group [15], that must in the basic form for catalytic activity of the enzyme from pig liver. The results shown in Fig. 1 suggest that a group with a pK value in this region may be involved in the activity of the ox brain enzyme.

Like the enzyme from ox heart [14], that from ox brain has been shown to act with the magnesium complex of isocitrate as the true substrate (Willson & Tipton, unpublished results). However, at the high levels of magnesium ions present in the assay, application of the computer analysis described by Storer and Cornish-Bowden [16] indicated that the observed sigmoid effects were not due to concentration-dependent changes in the relative proportions of the complexed and free species.

The failure of the enzyme to obey hyperbolic kinetics with isocitrate as the substrate at low concentrations of NAD^+ makes difficult any simple mechanistic interpretation of the kinetic studies at pH 6.5. The kinetic patterns observed at pH 6.5 with NAD^+ as the variable substrate were similar to those previously observed at pH 7.5 [1] in indicating that the apparent maximum velocity of the reaction is unaffected by changes in the isocitrate concentration. For a simple bisubstrate reaction this could be indicative of a mechanism in which isocitrate bound to the free enzyme under conditions of thermodynamic equilibrium before the binding of NAD^+ . Such an ordered binding is, however, not consistent with the observation that low concentrations of NAD^+ can bind to the free enzyme in order to displace it from an NAD -agarose affinity resin [6]. Competitive inhibition by NADH with respect to NAD^+ might also suggest that both compounds could bind to the free enzyme although a Theorell-Chance mechanism with NAD^+ as the second substrate would also give such an inhibition pattern. The apparently linear double-reciprocal plots obtained at higher NAD^+ concentrations with isocitrate as the substrate (Fig. 6) also

suggested that the apparent maximum velocity was insensitive to changes in the concentration of NAD^+ in this range. Such symmetrical patterns of primary double-reciprocal plots could occur if the dissociation constants of the substrates from binary complexes with the free enzyme were very much larger than their respective Michaelis constants.

The ability of NADH to induce a sigmoid response to isocitrate at pH 6.5 in the presence of relatively high concentrations of NAD^+ suggests that these two co-enzymes, although apparently competing for the same site on the enzyme, do not have the same effect on its allosteric response. A possible explanation for this could be that the observed behaviour results from complexities in the steady-state mechanism obeyed rather than from true cooperativity. If this were the case, however, a random-order mechanism would probably be unlikely since the inequalities necessary to produce a sigmoid response to isocitrate might also be expected to produce complex behaviour when the NAD^+ concentration was varied. In addition the sigmoid behaviour with isocitrate as the varied substrate would be expected to be lost at very high concentrations of NAD^+ but, although this is observed at pH 6.5, no such behaviour is observed at pH 7.5 [17]. A mnemonical mechanism [18] could, perhaps explain the behaviour, although no hysteretic behaviour in the progress curves was observed under any of the assay conditions used in this paper.

Plaut et al. [14] have proposed that the sigmoid behaviour with isocitrate as the substrate is due to cooperative interactions between subunits affecting the binding affinity. If this were the case the hyperbolic response to NAD^+ concentration and its ability to affect the Hill coefficient at pH 6.5 would be difficult to reconcile with a concerted model of cooperativity (see [19, 20] for example) and a sequential model (see [21]) would seem to be more likely. Direct studies of equilibrium saturation curves for substrate

and effector binding are necessary to resolve the possibilities.

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